

Assignment 04

1. Now write a code to display two paddles (no need to display the ball). One on the left corner and another one on the right corner.

[Use this Canva project as a reference take the help of guides for the boundary conditions](#)

Now map the left paddles y axis moment to buttons P17 (UP) and P18 (DOWN).

Right paddles y axis moment to buttons M18 (UP) and M17 (DOWN).

Make sure the y axis moment stops when it makes contact with the boundary.

2. Now taking everything to the next level.

You did code for the ball bouncing on the boundaries and also for the paddles. Now combine these two and write a logic for the basic ping pong game. No other improvements needed. Just the ball in red colour bouncing and paddles in blue. The ball only collides with the paddles when it comes to the side walls if the paddle misses the shot, the game should restart. Try writing this with just if conditions. (Hint: Just write a top module and copy paste both the ball bouncing and the paddles code and just manipulate the boundary conditions)

3. Now convert the previous 2nd question and improve the code by coding it in the form of a Finite state machine.